BC-48 Battery Charger: Service Manual

Document	Revision	Effective	Owner
NG-MAN-BC48	C	2025-03-01	Northgate Equipment Co.

1. Overview

The Northgate BC-48 is a high-frequency charger for 48V lead-acid and lithium-ion forklift batteries. It is networked over Ethernet and sends a heartbeat with temperature and power every 15 minutes. Most charging happens overnight (10 PM to 6 AM) and during the midday break.

2. Specifications

Item	Value
Output	48V DC, up to 250 A
Input	480V / 3 ph, 18 A
Max output power	14 kW
Network	Ethernet, heartbeat every 15 min

3. Normal Operating Ranges

Measurement	Normal range	Alarm limit
Cabinet temperature	80F to 110F	Above 130F (C-20)
Power draw while charging	10 to 14 kW	-
Power draw idle	Below 1 kW	-
Heartbeat	Every 15 min	None for 15 min (C-01)

Fault Code Reference

Code	Name	Trigger condition	Severity
C-01	Communication loss	No heartbeat for 15 minutes. The charger logs C-01 when it reconnects.	Medium
C-12	Ground fault	Leakage current above 30 mA.	High
C-20	Over-temperature	Cabinet temperature above 130F.	Medium
C-33	Battery not accepting charge	Output current below 10 A for 20 minutes at the start of a cycle.	Low

Troubleshooting by Fault Code

C-01: Communication loss

Trigger: No heartbeat for 15 minutes. The charger logs C-01 when it reconnects.

Likely causes, most common first:

- Network switch or cable fault at the dock.
- Charger powered off at the disconnect.
- Controller board lockup.

Corrective actions:

1. Check that the charger display is on. If dark, check the disconnect and breaker.
2. Check the Ethernet link light. Reseat or replace the patch cable.
3. Power cycle the controller (disconnect off for 60 seconds).
4. Check whether any batteries missed their overnight charge and tell the shift lead.

C-12: Ground fault

Trigger: Leakage current above 30 mA.

Likely causes, most common first:

- Damaged output cable.
- Battery case contamination.

Corrective actions:

1. Tag the charger out of service.
2. Inspect the output cable and connector.
3. Clean the battery top.

C-20: Over-temperature

Trigger: Cabinet temperature above 130F.

Likely causes, most common first:

- Blocked air filter.
- Failed cooling fan.

- Charger installed in direct sun.

Corrective actions:

1. Clean the intake filter.
2. Check the fan.
3. Provide 12 in clearance on all sides.

C-33: Battery not accepting charge

Trigger: Output current below 10 A for 20 minutes at the start of a cycle.

Likely causes, most common first:

- Sulfated or failed battery.
- Loose connector.

Corrective actions:

1. Check the connector.
2. Send the battery for capacity testing.

Preventive Maintenance

Task	Interval
Inspect output cable and connector	Monthly
Clean intake filter	Monthly
Check network connection	Quarterly

WARNING: Apply lockout/tagout per SOP-OPS-001 before opening any electrical enclosure or touching moving parts. Only authorized technicians may perform these steps.